ML-Based Cryptanalysis of Classical Ciphers

(Caesar, Affine & Vigenere) using Language Feature Extraction

B. Tech. Computer Science & Engineering
CSD334 MINI PROJECT 2026

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C E R T I F I C A T E

This is to certify that this report titled **ML-Based Cryptanalysis of Classical Ciphers (Caesar, Affine & Vigenere) using Language Feature Extraction** is a bona fide record of the work done by

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Sixth semester B. Tech. Computer Science and Engineering

Sixth Semester B. Tech. Computer Science & Engineering students, for the course work in **CSD 334 Mini Project**. The project titled **"ML-Based Cryptanalysis of Classical Ciphers"** focuses on automatic identification and decryption of classical ciphers using statistical analysis and machine learning techniques. This work is submitted in partial fulfillment of the requirements for the award of the degree of B. Tech. Computer Science and Engineering of **APJ Abdul Kalam Technological University**.

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ABSTRACT

This project presents an automatic system for detecting and decrypting classical ciphertexts without prior knowledge of the key or cipher type. The proposed tool focuses on widely used monoalphabetic and polyalphabetic schemes such as Caesar, Affine, and Vigenere ciphers, where ciphertext-only attack scenarios are common in both academic exercises and introductory cryptanalysis studies. The system first normalizes the input text and then computes language-driven statistical features including letter-frequency distribution, entropy, and the Index of Coincidence. These features are used to characterize the ciphertext and support machine learning based cipher identification.

After feature extraction, a lightweight classifier predicts the most likely cipher family and routes the input to the corresponding cryptanalytic module. For Caesar and Affine ciphers, exhaustive key search combined with chi-square scoring against English letter frequencies is used to recover the plaintext. For Vigenere ciphertexts, key-length estimation methods such as the Friedman test and Kasiski examination are applied, followed by frequency-based key recovery. Candidate plaintexts are ranked using English-likeness measures so that the most probable output is selected.

The project combines feature engineering, machine learning, and classical statistical attacks into a single automated workflow. In addition to the analytical pipeline, the system is designed as an educational platform that helps users understand how ciphertext preprocessing, feature extraction, cipher classification, and decryption strategies work together. The resulting framework is efficient, extensible, and suitable for mini-project level study in applied cryptanalysis.

Contents

1 Introduction **1**
1.1 Proposed Project 1
1.1.1 Problem Statement 1
1.1.2 Proposed Solution 1
2 Report of Preparatory Work **2**
2.1 Literature Survey Report 2
2.2 Tables 8
2.3 Feasibility 9
2.3.1 Technical Feasibility 9
2.3.2 Operational Feasibility 9
2.3.3 Economic Feasibility 9
3 Software Requirements Specification **10**
3.1 Introduction 10
3.2 Functional Requirements 10
3.2.1 Input Preprocessing 10
3.2.2 Feature Extraction 10
3.2.3 Cipher Identification 10
3.2.4 Automated Decryption 10
3.2.5 Result Presentation 11
3.3 Non-Functional Requirements 11
3.3.1 Performance 11
3.3.2 Accuracy and Reliability 11
3.3.3 Usability 11
3.3.4 Maintainability 11
3.3.5 Portability 11
4 Software Design Description **12**
4.1 System Design 12
4.1.1 Use Case Diagram 12
4.1.2 State Diagram 13
4.1.3 Level 0 Data Flow Diagram 13
4.1.4 Level 1 Data Flow Diagram 14
4.1.5 Activity Diagram 15
4.2 Module Description 15
4.3 Hardware and Software Requirements 16
4.3.1 Hardware Requirements 16
4.3.2 Software Requirements 16
5 Testing **17**
5.1 Unit Testing 17

5.1.1 Input Processing 17
5.1.2 Feature Extraction 17
5.1.3 Cipher Classification 17
5.1.4 Decryption Module 17
5.2 Integration Testing 17
5.2.1 End-to-End Workflow 17
5.2.2 Accuracy Verification 17
5.2.3 Performance Testing 17
5.3 Application Screenshots 18
6 Conclusion 22
7 Future Scope 23
References 24

List of Figures

2.1 Summary of Publications 8
4.1 Use Case Diagram 12
4.2 State Diagram 13
4.3 Level 0 Data Flow Diagram 13
4.4 Level 1 Data Flow Diagram 14
4.5 Activity Diagram 15
5.1 Krypto encryption interface 18
5.2 Krypto decryption interface 18
5.3 Caesar cipher encryption result 18
5.4 Vigenere cipher encryption result 19
5.5 Affine cipher encryption result 19
5.6 Caesar cipher decryption output 19
5.7 Caesar cipher statistics and prediction visualization 20
5.8 Vigenere cipher decryption output 20
5.9 Vigenere cipher statistics and feature radar chart 20
5.10 Affine cipher decryption output 21
5.11 Affine cipher statistics and prediction visualization 21

Chapter 1

Introduction

Classical cryptography remains an important foundation for understanding modern secure communication systems. Although ciphers such as Caesar, Affine, and Vigenere are no longer secure for real-world confidential communication, they are still widely used in education to explain substitution techniques, key-based transformations, and basic attack models. Manual cryptanalysis of such ciphers is time consuming because the analyst must first guess the cipher family and then apply the correct decryption strategy.

1.1 Proposed Project

1.1.1 Problem Statement

Given only a ciphertext sample, it is often difficult to determine whether the text was produced using a monoalphabetic or polyalphabetic method. A wrong assumption about the cipher type leads to ineffective attacks and wasted effort. The project addresses the problem of automatically identifying the most likely classical cipher type and decrypting the message without prior knowledge of the encryption key.

1.1.2 Proposed Solution

The proposed solution is an ML-assisted cryptanalysis system for Caesar, Affine, and Vigenere ciphers. The system accepts ciphertext as input, preprocesses it, extracts language and pattern based features, predicts the cipher family using a machine learning classifier, and then applies the most suitable attack technique. The full workflow includes:

input normalization by removing irrelevant symbols and standardizing case,
feature extraction using frequency statistics and Index of Coincidence,
cipher identification using a trained lightweight ML model,
targeted decryption using brute-force or key-recovery methods, and
ranking candidate plaintexts using English-likeness scoring.

Chapter 2

Report of Preparatory Work

2.1 Literature Survey Report

"A Massive Machine-Learning Approach for Classical Cipher Type Detection Using Feature Engineering" - Leierzopf, E., Kopal, N., Esslinger, B., Lampesberger, H., & Hermann, E. [1]

This paper proposes a highly scalable, machine-learning-driven approach to automatically classify classical cipher types, moving beyond manual cryptanalysis. Recognizing the challenges historical ciphers pose in automated systems, the researchers developed a methodology anchored in engineered statistical features. By analyzing text patterns and language characteristics without relying on the exact cipher class, the study streamlines the cryptanalytic pipeline. This foundational piece shows how classical cryptography intersects with data science to accelerate decryption.

Implementation Details

The technical contribution involves training and systematically evaluating machine learning models, including Feed-Forward Neural Networks, Decision Trees, Random Forests, and Naive Bayes Networks. The authors constructed a massive dataset comprising 56 distinct cipher types from the American Cryptogram Association (ACA). Through rigorous feature engineering—incorporating the Index of Coincidence, entropy vectors, and n-gram analysis—the highest-performing model attained an impressive 80.24% recognition accuracy. This architecture was then packaged into a software tool known as the Neural Cipher Identifier (NCID).

Advantages:

The architecture fundamentally supports versatile cipher detection across 56 distinct classical cipher variants, making it a stellar reference for broad cryptanalysis tools. Furthermore, achieving a recognition accuracy margin that exceeds 80% represents a strong machine learning baseline. Such high reliability minimizes the need for computationally heavy brute-force trial and error across multiple decryption algorithms.

Disadvantages:

Despite robust performance, the system heavily relies on exceptionally diverse training datasets, demanding substantial initial data-generation overhead. Consequently, operational routines require high computational power and memory, limiting immediate portability to low-resource environments.

"A Study on some modified Classical Ciphers for Secure Crypto-System" - Sravani, J., Chandrasekhar, A., & Pragathi, C. [2]

This paper investigates the deliberate modernization of historical cryptography by proposing a modified multiple-encryption scheme that combines Caesar and Affine ciphers over Unicode characters. The research seeks to demonstrate that while classical ciphers are traditionally obsolete against modern computing power, restructuring their operational domain can drastically elevate their cryptographic resilience. It explores how ancient algorithms can be retrofitted to withstand standard foundational attacks effectively.

Implementation Details

The technical methodology applies a sequential, double-layered encryption process whereby the plaintext is first subjected to a Caesar shift and subsequently scrambled using Affine algebraic transformations. The authors eschewed the traditional 26-letter English alphabet framework and integrated a significantly larger Unicode character set. This mathematical expansion multiplies the available key space. By operating across a larger matrix of potential characters, the system creates a high-entropy ciphertext that naturally resists standard frequency analysis.

Advantages:

The standout advantage of this methodology is its innovative approach to reinforcing baseline algorithmic security simply by combining Caesar and Affine principles into a unified, stronger sequence. The deliberate implementation of a massive Unicode key space effectively cripples simple brute-force vulnerability checks and limits simple attacks.

Disadvantages:

While the encryption strategy is notably creative, the research is entirely isolated to novel encryption and decryption techniques, completely sidelining the subject of automated cipher detection or unknown ciphertext analysis.

"Identifying and Deciphering Ciphers: Leveraging Machine Learning for Streamlined Decryption" - Kosta, S., Kujur, S., & Lokhandwala, T. [3]

This paper introduces a streamlined strategy exploring the use of modern machine learning techniques to automate both the identification and the subsequent decryption phases for classical ciphers. Focusing heavily on foundational schemes such as Caesar, simple substitution, and Vigenère ciphers, the authors aim to bridge the gap between abstract classification and actionable cryptanalysis. By transitioning from traditional, highly manual cryptanalysis to an automated pipeline, the research underlines the growing necessity of intelligent computational tools in overcoming historical cryptography barriers, providing a theoretical blueprint for integrating identification with automated key-search operations.

Implementation Details

The research extensively reviews traditional cryptanalysis methods—ranging from basic frequency analysis to known-plaintext attacks—and critically analyzes their operational bottlenecks. Building on this theoretical groundwork, it proposes various machine learning-based approaches precisely tuned to improve processing speed, operational accuracy, and overall workflow seamlessness. The core structure centers on feeding localized ciphertext data into learning models that determine the encryption mechanism. The study elaborates on the effectiveness, potential cybersecurity applications, and current limitations of automated cryptanalysis systems, serving as an architectural guide for building integrated tools.

Advantages:

The primary strength of this research lies in its strong conceptual alignment with practical ML-based detection specifically for classical ciphers, making it highly relevant to projects focusing on specific encryption mechanisms. Unlike works that exclusively address cipher classification, this paper significantly extends its scope by encouraging the automation of both the identification phase and the final decryption algorithm. This end-to-end approach simplifies the cryptographic pipeline and dramatically reduces the requirement for human intervention during ciphertext analysis.

Disadvantages:

While structurally sound in theory, the publication remains largely theoretical and heavily review-based. It offers limited concrete technical implementation details regarding optimal feature selection or exact neural layer configurations. Furthermore, the absence of detailed empirical performance metrics, extensive validation datasets, or rigorous accuracy comparisons against existing methodologies makes it difficult to directly quantify the proposed system's reliability in real-world, noisy ciphertext scenarios.

"Classification and Identification of Classical Cipher Type using Artificial Neural Networks" - Abd, A.J., & Al-Janabi, S.T.F. [4]

This scholarly article proposes a powerful, multi-level Artificial Neural Network (ANN) model specifically tailored for classifying and identifying varied classical cipher types. Recognizing the mathematical patterns inherent in historical cryptography, the research capitalizes on the deep learning capabilities of neural networks to distinguish between complex ciphertext structures. By interpreting encrypted text as feature-rich data streams, the authors demonstrate how deep hierarchical learning can be successfully deployed to categorize historical ciphers with extraordinary precision, minimizing human analytical bias.

Implementation Details

The implemented system adopts a multi-tiered diagnostic approach. In its first level, it heuristically categorizes incoming ciphers into broad cryptographic classes: transposition systems, substitution systems, and composite combination systems. After establishing the primary cryptographic family, the network funnels the data into specialized secondary layers configured to further identify specific cipher types, such as the Vigenère, Playfair, and Hill ciphers. Through rigorous training and validation phases utilizing diverse encrypted datasets, the proposed multi-level ANN achieved a remarkably high classification accuracy, registering at approximately 99% during final testing phases.

Advantages:

The core advantage of this study is its highly organized, structured multi-level classification approach, which logically breaks down complex cryptanalysis into manageable, sequential prediction tasks. This modularity ensures the system is both scalable and highly organized. Moreover, the exceptionally high reported accuracy rate of 99% in cipher identification heavily reinforces the viability and effectiveness of applying artificial neural networks to solve deterministic cryptography problems.

Disadvantages:

Despite its exceptional classification accuracy, the system primarily focuses exclusively on the identification phase, lacking an integrated mechanism for full decryption and plaintext recovery. Developing and refining this multi-level ANN requires an extensive, heavily annotated training dataset and exhaustive hyperparameter tuning, which consumes vast computational resources. Implementing such a nuanced neural architecture can be disproportionately challenging for standalone offline projects lacking dedicated GPU hardware.

"Classifying Classic Ciphers using Machine Learning" - Krishna, N.R.R. [5]

This analytical research delves specifically into identifying fundamental classical ciphers —including Simple Substitution, Vigenère, Playfair, and Columnar Transposition— using a diverse array of machine learning techniques. By treating cryptanalysis as a distinct pattern recognition and classification problem, the study comparatively examines different algorithmic workflows to ascertain the most efficient mathematical pipeline. The research serves as a comparative benchmark, contrasting traditional statistical learning mechanisms against deep convolutional architectures to map out the future of automated cryptographic analysis.

Implementation Details

The investigation methodically tested three distinct computational approaches. The first approach utilized Support Vector Machines (SVM) trained directly on raw ciphertext arrays. The second, more advanced approach, transformed the problem into an image classification task by training Convolutional Neural Networks (CNN) on visually rendered Hidden Markov Model (HMM) generated images. The final approach trained an SVM using refined HMM feature vectors. By running these parallel experimental paths, the results provided a comprehensive comparison detailing the computational effectiveness, training time, and predictive accuracy of each distinct modeling paradigm against classical ciphers.

Advantages:

This research is highly valuable as it systematically compares multiple distinct machine learning techniques for cipher classification side-by-side, offering rare empirical insight into algorithm selection. By intentionally experimenting with structured feature extraction using hybrid models—such as the fusion of HMM capabilities with CNN or SVM classification—it showcases innovative problem-solving in data formatting. These specific architectural insights are exceptionally useful for balancing accuracy against processing constraints.

Disadvantages:

The study's scope is entirely restricted to classification tasks, providing no architectural guidance for actual key extraction or full decryption processes. Furthermore, the combination of Hidden Markov Models with Convolutional Neural Networks forms an extremely computationally intensive pipeline. This high processing requirement demands advanced hardware and significant runtime, rendering the proposed HMM + CNN methodology largely impractical for lightweight, rapid-response educational cryptanalysis tools.

"Various Ciphers in Classical Cryptography" - Deepthi, D.V.V., Benny, B.H., & Sreenu, K. [6]

This fundamentally descriptive paper comprehensively dissects and analyzes a wide variety of classical cryptographic techniques, acting as a historical and mathematical survey of the field. It methodically covers foundational algorithms including the Caesar cipher, Playfair, Hill cipher, various Polyalphabetic schemes, the Rail Fence sequence, and Columnar Transposition ciphers. The primary objective is to build a structured, unified understanding of historical encryption logic, laying out the theoretical foundation necessary before progressing to modern, mathematically dense cryptographic standards or engaging in programmatic cryptanalysis.

Implementation Details

The study primarily functions through a rigorous comparative methodology. It systematically categorizes and compares substitution and transposition encryption methods based on distinct operational criteria: core key type requirements, fixed versus variable block sizes, susceptibility to specific attack vectors (like known-plaintext or ciphertext-only attacks), and their overarching algorithmic strength. The resulting matrix provides a solid foundational understanding of classical encryption behaviors, security properties, and inherent vulnerabilities, effectively cataloging the mechanical logic behind ancient cryptography.

Advantages:

The primary benefit of this work lies in its exceptional clarity; it provides an easily digestible, direct comparison of diverse classical cipher techniques from a purely mechanical perspective. By thoroughly covering monoalphabetic schemes like Caesar and explicitly detailing Polyalphabetic systems closely related to the Vigenère cipher, it grants essential theoretical context. This context is highly applicable when designing the deterministic mathematical boundaries and parameters for any custom cryptanalytic software.

Disadvantages:

As a largely theoretical and survey-oriented analysis, the text lacks any discussion regarding software implementation, algorithmic automation, or computational framework design. Furthermore, it strictly remains within the bounds of classical cryptanalysis theory and entirely omits any modern discourse on automated cipher detection, statistical feature extraction, or the application of machine learning approaches, limiting its utility specifically to foundational mathematical reference.

2.2 Tables

Publication Year	Paper	Methodology	Disadvantage
2021	A Massive Machine-Learning Approach for Classical Cipher Type Detection Using Feature Engineering	Engineered statistical features, Multiple classifiers (FFNN, DT, RF, NBN)	Requires large training data and high computational power
2021	A Study on some modified Classical Ciphers for Secure Crypto-System	Double encryption combining Caesar and Affine over Unicode characters	Still based on classical methods, no ML-based analysis
2020	Identifying and Deciphering Ciphers: Leveraging Machine Learning for Streamlined Decryption	ML techniques for automated cipher identification and decryption	Largely theoretical and review-based
2019	Classification and Identification of Classical Cipher Type using Artificial Neural Networks	Multi-level Artificial Neural Network (ANN) model	Requires extensive training data and ANN tuning
2019	Classifying Classic Ciphers using Machine Learning	SVM trained on ciphertext, CNN on HMM images, SVM on HMM features	Computationally intensive (HMM + CNN pipeline)
2019	Various Ciphers in Classical Cryptography	Comparison of substitution and transposition methods	Theoretical analysis only, no automation

Table 2.1: Summary of Publications

2.3 Feasibility

2.3.1 Technical Feasibility

The project is technically feasible because it combines standard programming techniques with well-known statistical methods from classical cryptanalysis. Feature extraction, brute-force search, chi-square scoring, and key-length estimation can all be implemented with moderate computational resources. The ML component is lightweight because the supported cipher set is intentionally limited.

2.3.2 Operational Feasibility

The system is useful for students, beginners in cryptography, and instructors who want a simple tool to analyze ciphertext samples. It reduces manual trial-and-error and improves understanding by showing how the system moves from ciphertext to feature extraction, classification, and decryption.

2.3.3 Economic Feasibility

The project does not require specialized hardware or paid services. Development can be carried out using common programming tools and open learning resources, making it suitable for an academic mini-project setting.

Chapter 3

Software Requirements Specification

3.1 Introduction

This Software Requirements Specification describes the functional and non-functional requirements of the project ML-Based Cryptanalysis of Classical Ciphers (Caesar, Affine & Vigenere) using Language Feature Extraction. The document captures the expected behavior of the system, supported workflows, major constraints, and user-facing features.

The system accepts ciphertext as input and performs automatic cipher analysis for Caesar, Affine, and Vigenere ciphers. It preprocesses the text, extracts statistical language features, predicts the cipher family through an ML classifier, and applies the relevant decryption attack. The output includes the predicted cipher type, likely plaintext, and supporting information such as key guesses or ranking scores.

3.2 Functional Requirements

3.2.1 Input Preprocessing

The system shall accept ciphertext input from the user.
The system shall convert input to uppercase and remove spaces, digits, and special characters as required for analysis.

3.2.2 Feature Extraction

The system shall compute statistical features including letter-frequency distribution and Index of Coincidence.
The system may compute entropy, repeating patterns, autocorrelation, and other supporting indicators.

3.2.3 Cipher Identification

The system shall use extracted features to predict whether the ciphertext belongs to Caesar, Affine, or Vigenere cipher families.
The system shall select an attack strategy based on the predicted cipher type.

3.2.4 Automated Decryption

The system shall run brute-force attacks for Caesar and Affine ciphers.
The system shall estimate key length and recover the key for Vigenere cipher.
The system shall generate and score plaintext candidates.

3.2.5 Result Presentation

The system shall display the predicted cipher type.
The system shall present the most probable decrypted plaintext.
The system shall show useful metadata such as key guesses, scores, or key length estimates where applicable.

3.3 Non-Functional Requirements

3.3.1 Performance

The system should produce results within a short time for typical ciphertext lengths used in academic examples.
Feature extraction and scoring should remain efficient enough for interactive use.

3.3.2 Accuracy and Reliability

Cipher identification should be sufficiently reliable for the supported three-cipher scope.
Decryption ranking should prioritize English-like plaintext outputs.

3.3.3 Usability

The interface should be easy to use for students with basic cryptography knowledge.
Output should be readable and should clearly distinguish cipher prediction from final plaintext.

3.3.4 Maintainability

The system should be modular so that new cipher families can be added in future.
Preprocessing, classification, and attack modules should remain loosely coupled.

3.3.5 Portability

The project should run on common academic computing environments without specialized setup.

Chapter 4

Software Design Description

4.1 System Design

The system is designed to provide an automated workflow for identifying and decrypting classical ciphers. It integrates preprocessing, feature extraction, machine learning based classification, and cipher-specific decryption modules into a single pipeline.

4.1.1 Use Case Diagram

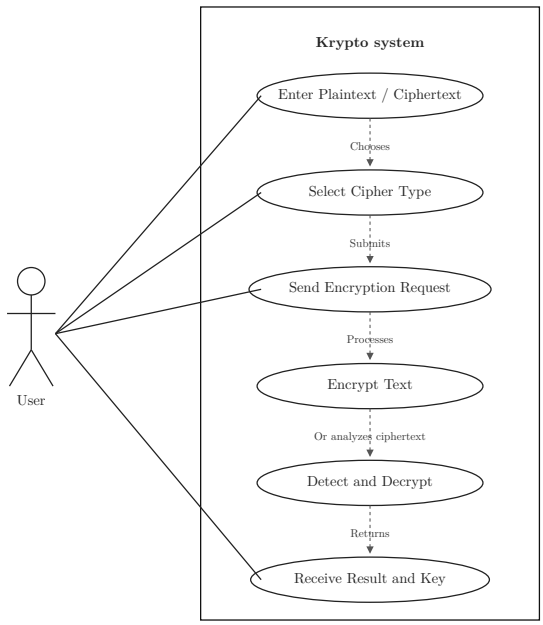

Figure 4.1: Use Case Diagram

The use case diagram represents the interaction between the user and the system. The user provides ciphertext input, and the system processes it through feature extraction, cipher identification, and decryption. The final plaintext output is returned to the user.

4.1.2 State Diagram

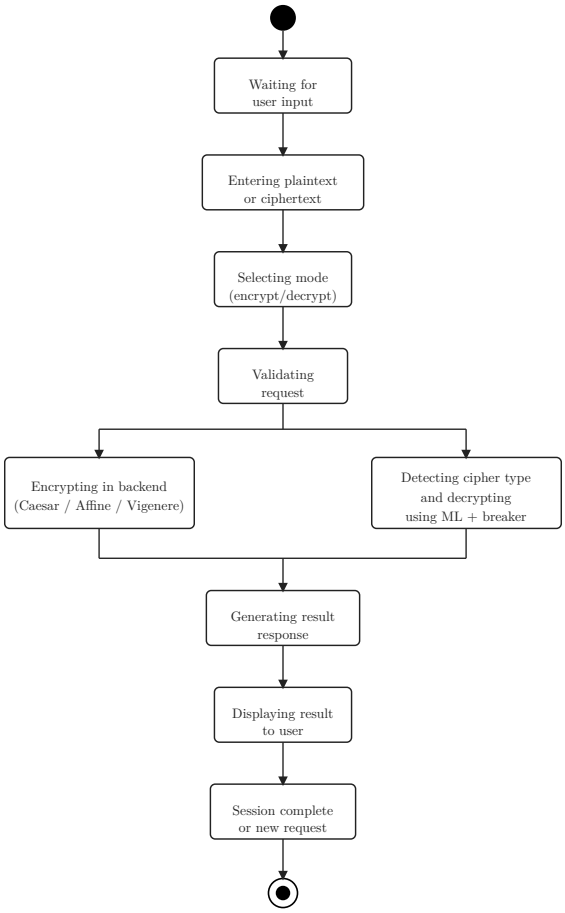

Figure 4.2: State Diagram

The state diagram illustrates the system workflow. The system starts in an idle state, accepts ciphertext input, processes it, identifies the cipher, performs decryption, and outputs the result before returning to the initial state.

4.1.3 Level 0 Data Flow Diagram

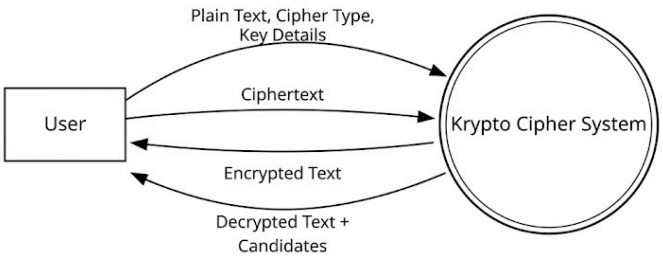

Figure 4.3: Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram provides a high-level overview of the system. It shows how ciphertext input flows into the system, is processed through analysis modules, and produces decrypted output.

4.1.4 Level 1 Data Flow Diagram

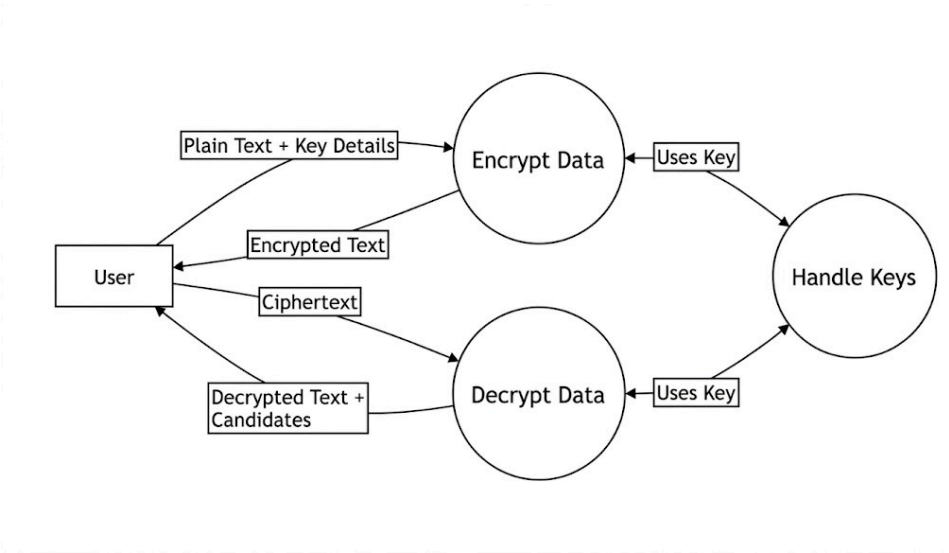

Figure 4.4: Level 1 Data Flow Diagram

The Level 1 DFD provides a detailed view of internal processing. It shows feature extraction, classification, and decryption steps as separate components working together.

4.1.5 Activity Diagram

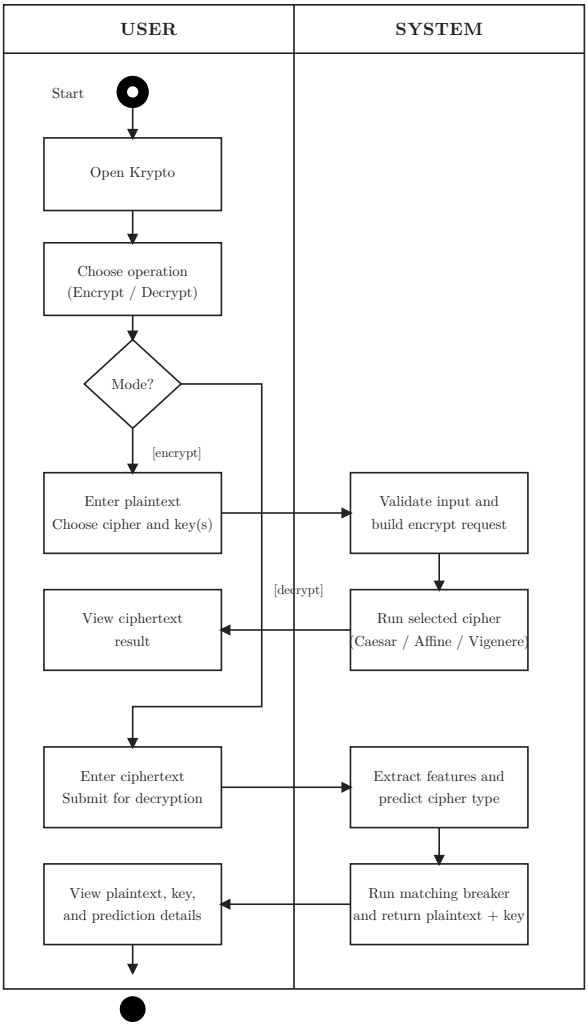

Figure 4.5: Activity Diagram

The activity diagram represents the step-by-step workflow of the system. It includes input processing, feature extraction, classification, decryption, and output generation.

4.2 Module Description

The system is divided into multiple modules:

Input Processing Module: Handles ciphertext normalization and cleaning.
Feature Extraction Module: Computes statistical features such as frequency and Index of Coincidence.
Classification Module: Predicts cipher type using machine learning.
Decryption Module: Applies cipher-specific attacks.
Output Module: Displays decrypted plaintext and analysis results.

4.3 Hardware and Software Requirements

4.3.1 Hardware Requirements

Standard personal computer or laptop
Minimum 8 GB RAM
Multi-core processor recommended

4.3.2 Software Requirements

Operating system: Windows, Linux, or macOS
Programming language: Python
Libraries for machine learning and text processing
Development tools and environment for implementation

Chapter 5

Testing

The system was tested to ensure that each module performs correctly and that the complete workflow operates as expected.

5.1 Unit Testing

5.1.1 Input Processing

The input preprocessing module was tested to ensure that ciphertext is correctly normalized by removing unwanted characters and standardizing case.

5.1.2 Feature Extraction

The feature extraction module was tested to verify correct computation of statistical features such as letter frequency, entropy, and Index of Coincidence.

5.1.3 Cipher Classification

The classification module was tested using sample ciphertexts to ensure accurate prediction of cipher types.

5.1.4 Decryption Module

The decryption modules for Caesar, Affine, and Vigenere ciphers were tested to ensure correct recovery of plaintext.

5.2 Integration Testing

5.2.1 End-to-End Workflow

The complete pipeline was tested by providing ciphertext input and verifying that the system correctly performs preprocessing, classification, and decryption.

5.2.2 Accuracy Verification

The decrypted outputs were compared against known plaintexts to evaluate correctness.

5.2.3 Performance Testing

The system was tested for execution time and responsiveness to ensure it performs efficiently for typical input sizes.

5.3 Application Screenshots

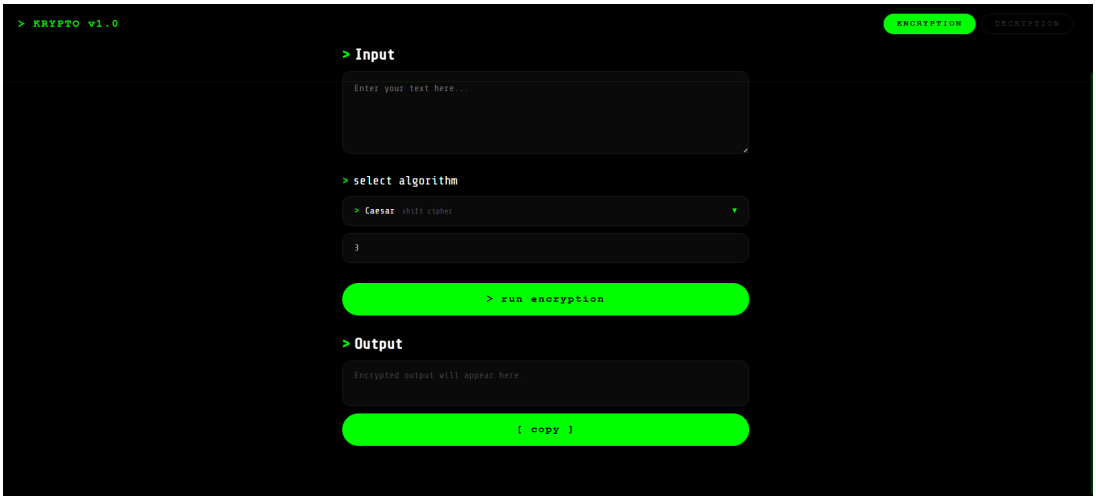

Figure 5.1: Krypto encryption interface

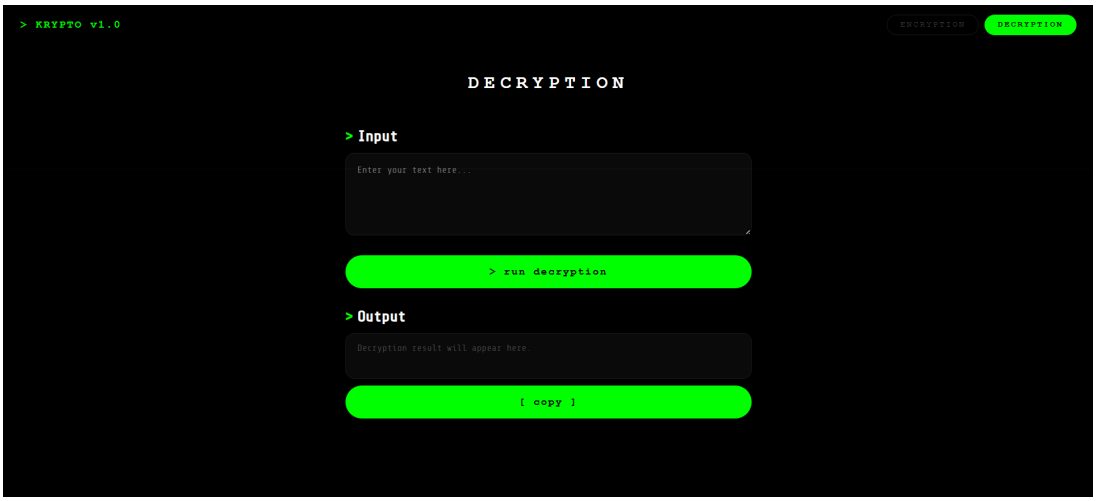

Figure 5.2: Krypto decryption interface

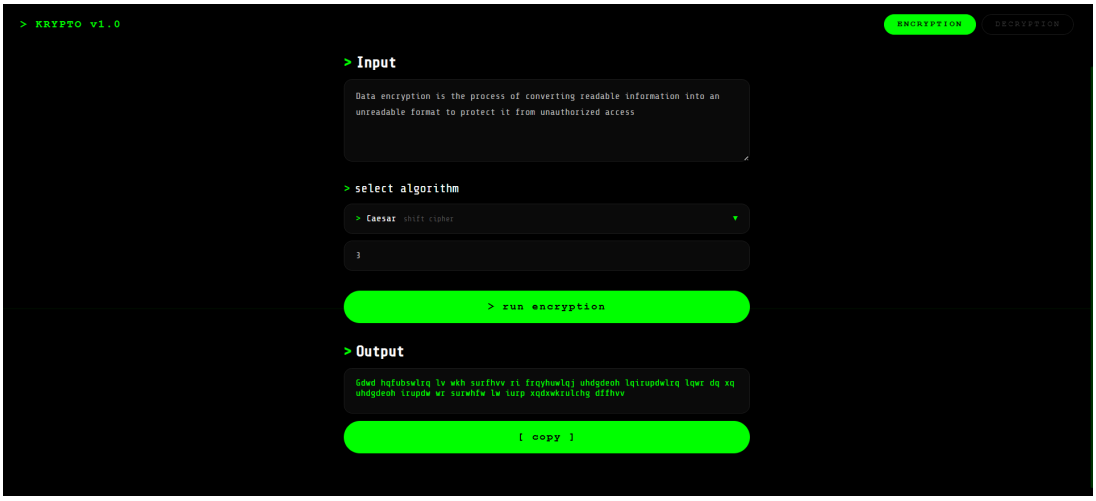

Figure 5.3: Caesar cipher encryption result

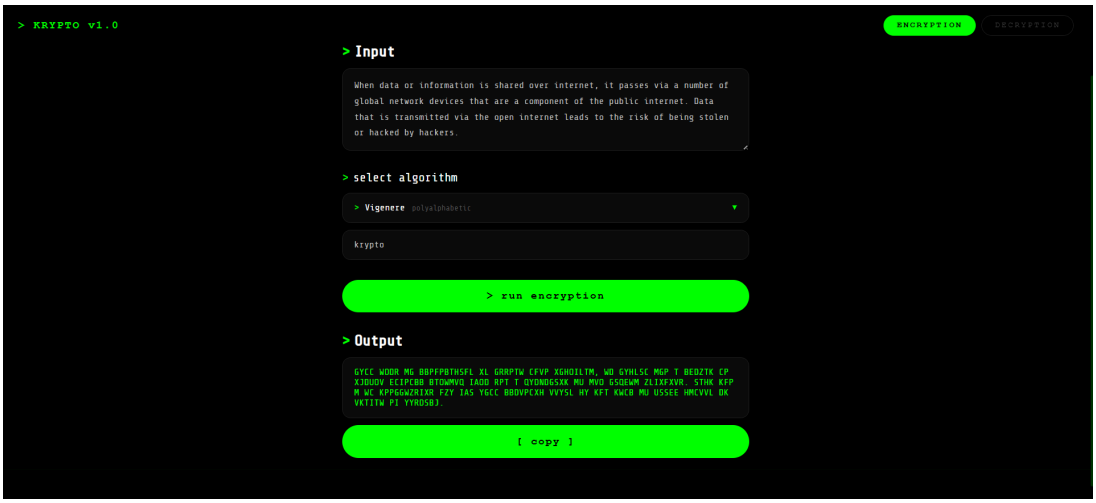

Figure 5.4: Vigenere cipher encryption result

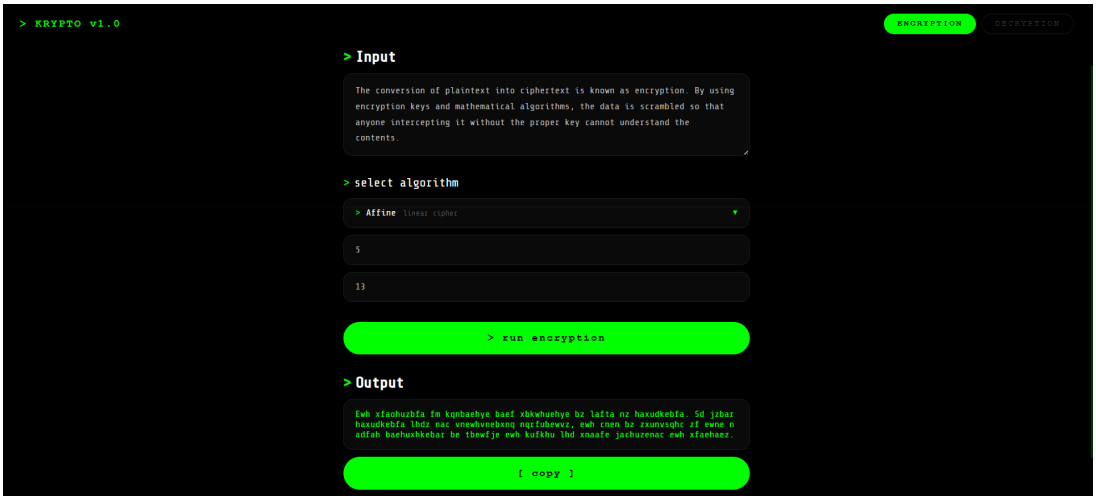

Figure 5.5: Affine cipher encryption result

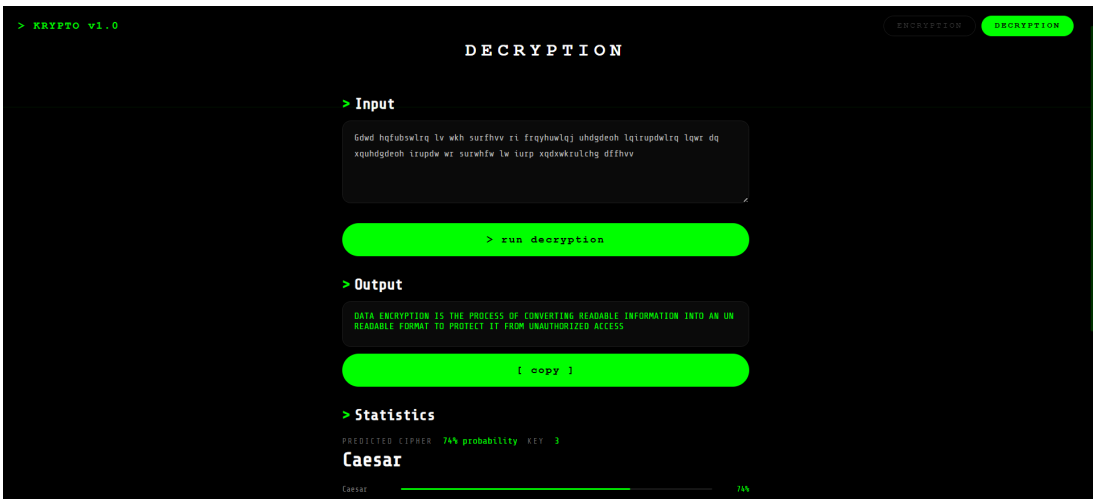

Figure 5.6: Caesar cipher decryption output

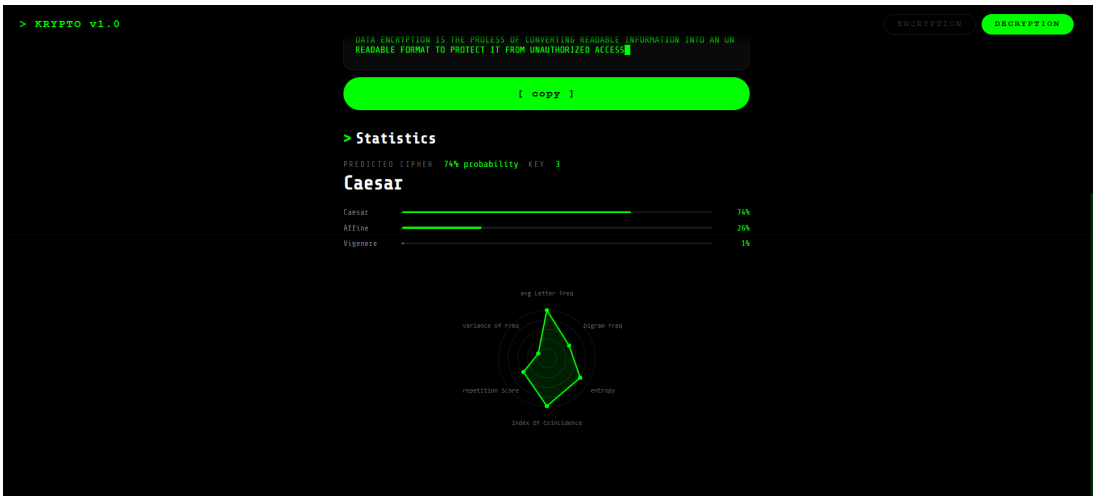

Figure 5.7: Caesar cipher statistics and prediction visualization

Figure 5.8: Vigenere cipher decryption output

Figure 5.9: Vigenere cipher statistics and feature radar chart

Figure 5.10: Affine cipher decryption output

Figure 5.11: Affine cipher statistics and prediction visualization

Chapter 6

Conclusion

This mini project presents a complete workflow for the automatic identification and decryption of classical ciphers using language feature extraction. By combining preprocessing, feature engineering, machine learning based cipher classification, and targeted cryptanalytic attacks, the system reduces the manual effort usually involved in ciphertext-only analysis.

The project successfully addresses three important classical ciphers: Caesar, Affine, and Vigenere. Monoalphabetic ciphers are handled through brute-force search and chi-square scoring, while the polyalphabetic pipeline uses key-length estimation and frequency-based recovery strategies. The classifier acts as the decision layer that connects feature extraction to the correct decryption module.

Beyond solving ciphertext samples, the project also serves as a learning tool. It demonstrates how statistical language properties can guide cipher detection and how traditional cryptanalytic ideas can be integrated with machine learning in a practical and modular way. The framework can be extended in future to support additional ciphers and more advanced scoring methods.

Chapter 7

Future Scope

The current system focuses on Caesar, Affine, and Vigenere ciphers. Future improvements can extend the framework to support additional classical ciphers such as Playfair, Hill, and substitution-based variants.

The accuracy of cipher identification can be improved by training on larger and more diverse datasets. Advanced machine learning models or deep learning approaches may further enhance classification performance.

The system can also be extended to support multiple languages instead of being limited to English-based statistical analysis. Improved scoring techniques such as n-gram models can enhance decryption accuracy.

A more interactive user interface can be developed to visualize intermediate steps such as feature extraction, classification confidence, and key recovery process for better educational understanding.

References

1. Leierzopf, E., Kopal, N., Esslinger, B., Lampesberger, H., & Hermann, E. "A Massive Machine-Learning Approach for Classical Cipher Type Detection Using Feature Engineering." (Neural Cipher Identifier - NCID).
2. Sravani, J., Chandrasekhar, A., & Pragathi, C. "A Study on some modified Classical Ciphers for Secure Crypto-System." Turkish Journal of Computer and Mathematics Education, Vol.12, No.6, 2021, pp. 5316-5319.
3. Kosta, S., Kujur, S., & Lokhandwala, T. "Identifying and Deciphering Ciphers: Leveraging Machine Learning for Streamlined Decryption." International Journal of Computer Engineering and Technology (IJCET), 11(3), 2020, pp. 68-72.
4. Abd, A.J., & Al-Janabi, S.T.F. "Classification and Identification of Classical Cipher Type using Artificial Neural Networks." Journal of Engineering and Applied Sciences, 14(11), 2019, pp. 3549-3556.
5. Krishna, N.R.R. "Classifying Classic Ciphers using Machine Learning." Master's Project, San José State University, 2019.
6. Deepthi, D.V.V., Benny, B.H., & Sreenu, K. "Various Ciphers in Classical Cryptography." Journal of Physics: Conference Series, Vol. 1228, 2019, 012014.